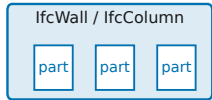


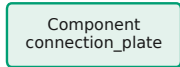
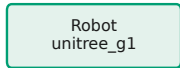
(a) IFC design model



IFC GUID
2A9nYhQvT5Q9B7m7X1pA01

design intent only

(b) assets pass alone



geometry, collision, mass OK

(c) composed scene fails



missing_workzone
payload



wrong_up_axis



unsupported
generated_friction

cross-asset faults (fault taxonomy)

(d) scene-level contract



- ✓ closed references
- ✓ provenance
- ✓ task preconditions